St. George's University
Grenada, West Indies

Basic Principles of Medicine 1

Module: Foundations To Medicine| Lecture 35

Histology of Muscle Tissue - Flipped Class

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Recommended Reading

Histology – A Text and Atlas
Pawlina 9th Edition

Chapter 11: Muscle Tissue Pg. 344-387

- https://sgu.idm.oclc.org/login?url=https://premiumbasicsciences-lwwhealthlibrary-com.sgu.idm.oclc.org/book.aspx?bookid=3290

Objectives

List the functions of muscle tissue.	SOM.MK.I.BPM1.1.FTM.3.HCB.0887
Describe the structure of thick myofilaments in striated muscles.	SOM.MK.I.BPM1.1.FTM.3.HCB.0888
Identify the proteints associated with the thick myofilament in striated muscles.	SOM.MK.I.BPM1.1.FTM.3.HCB.0889
Give the function(s) of the proteins associated with the thick myofilament in striated muscles	SOM.MK.I.BPM1.1.FTM.3.HCB.0890
Describe the structure of thin myofilaments in striated muscles	SOM.MK.I.BPM1.1.FTM.3.HCB.0891
Identify the proteins associated with the thin myofilament in striated muscles.	SOM.MK.I.BPM1.1.FTM.3.HCB.0892
Give the function(s) of the proteins associated with the thin myofilament in striated muscles	SOM.MK.I.BPM1.1.FTM.3.HCB.0893
Give the function of the sarcomere.	SOM.MK.I.BPM1.1.FTM.3.HCB.0894
Describe the microscopic structure of the sarcomere.	SOM.MK.I.BPM1.1.FTM.3.HCB.0895
Describe the sliding filament model of muscle contraction.	SOM.MK.I.BPM1.1.FTM.3.HCB.0896
Describe briefly the events of the actomyosin cross-bridge cycle.	SOM.MK.I.BPM1.1.FTM.3.HCB.0897
List the 2 types of striated muscle.	SOM.MK.I.BPM1.1.FTM.3.HCB.0898
Describe the contractile apparatus of striated muscle fibers.	SOM.MK.I.BPM1.1.FTM.3.HCB.0899
State the functions of the accessory proteins typically associated with striated muscle cells.	SOM.MK.I.BPM1.1.FTM.3.HCB.0900
Give the function(s) of skeletal muscle.	SOM.MK.I.BPM1.1.FTM.3.HCB.0901

Objectives

State the localization of skeletal muscle.	SOM.MK.I.BPM1.1.FTM.3.HCB.0902
Describe the microscopic structure of skeletal muscle.	SOM.MK.I.BPM1.1.FTM.3.HCB.0903
Describe the general organization of skeletal muscle including connective tissue, muscle fascicles, muscle fibers, myofibrils and myofilaments.	SOM.MK.I.BPM1.1.FTM.3.HCB.0904
Describe the structure, function and localization of the T-tubules and the sarcoplasmic reticulum in skeletal muscle.	SOM.MK.I.BPM1.1.FTM.3.HCB.0905
Describe the three types of skeletal muscle fibers.	SOM.MK.I.BPM1.1.FTM.3.HCB.0906
Discuss the regenerative capability of skeletal muscle.	SOM.MK.I.BPM1.1.FTM.3.HCB.0907
Describe the ultrastructural defects in muscular dystrophies including Duchenne muscular dystrophy.	SOM.MK.I.BPM1.1.FTM.3.HCB.0908
Give the function of the muscle spindle.	SOM.MK.I.BPM1.1.FTM.3.HCB.0909
Describe the histological structure of the muscle spindle.	SOM.MK.I.BPM1.1.FTM.3.HCB.0910
Give the function of the Golgi tendon organs.	SOM.MK.I.BPM1.1.FTM.3.HCB.0911
Describe the histological structure of the Golgi tendon organs.	SOM.MK.I.BPM1.1.FTM.3.HCB.0912
Give the function(s) of cardiac muscle.	SOM.MK.I.BPM1.1.FTM.3.HCB.0913
State the localization of cardiac muscle.	SOM.MK.I.BPM1.1.FTM.3.HCB.0914
Describe the microscopic structure of cardiac muscle.	SOM.MK.I.BPM1.1.FTM.3.HCB.0915

Objectives

Describe the general organization of cardiac muscle, including the structure and function of intercalated discs.	SOM.MK.I.BPM1.1.FTM.3.HCB.0916
Give the function(s) of Purkinje fibers	SOM.MK.I.BPM1.1.FTM.3.HCB.0917
Describe the microscopic structure Purkinje fibers.	SOM.MK.I.BPM1.1.FTM.3.HCB.0918
State the localization of Purkinje fibers.	SOM.MK.I.BPM1.1.FTM.3.HCB.0919
Describe the structure, function and localization of the T-tubules and the sarcoplasmic reticulum in cardiac muscle.	SOM.MK.I.BPM1.1.FTM.3.HCB.0920
Discuss the regenerative capability of cardiac muscle.	SOM.MK.I.BPM1.1.FTM.3.HCB.0921
Give the function(s) of smooth muscle.	SOM.MK.I.BPM1.1.FTM.3.HCB.0922
State the localization of smooth muscle.	SOM.MK.I.BPM1.1.FTM.3.HCB.0923
Describe the microscopic structure of smooth muscle.	SOM.MK.I.BPM1.1.FTM.3.HCB.0924
Describe the microscopic structure of the thick myofilaments in smooth muscles.	SOM.MK.I.BPM1.1.FTM.3.HCB.0925
Identify the proteins associated with the thick myofilament in smooth muscles.	SOM.MK.I.BPM1.1.FTM.3.HCB.0926
Give the function(s) of the proteins typically associated with the thick myofilament in smooth muscles	SOM.MK.I.BPM1.1.FTM.3.HCB.0927

Objectives

Describe the microscopic structure of thin myofilaments in smooth muscles	SOM.MK.I.BPM1.1.FTM.3.HCB.0928
Identify the proteins associated with the thin myofilament in smooth muscles.	SOM.MK.I.BPM1.1.FTM.3.HCB.0929
Give the function(s) of the proteins typically associated with the thin myofilament in smooth muscles.	SOM.MK.I.BPM1.1.FTM.3.HCB.0930
Describe the contractile apparatus of smooth muscle cells.	SOM.MK.I.BPM1.1.FTM.3.HCB.0931
Describe the regenerative capability of smooth muscle.	SOM.MK.I.BPM1.1.FTM.3.HCB.0932
State the function(s) of the accessory proteins usually associated with smooth muscle cells.	SOM.MK.I.BPM1.1.FTM.3.HCB.0933

Overview

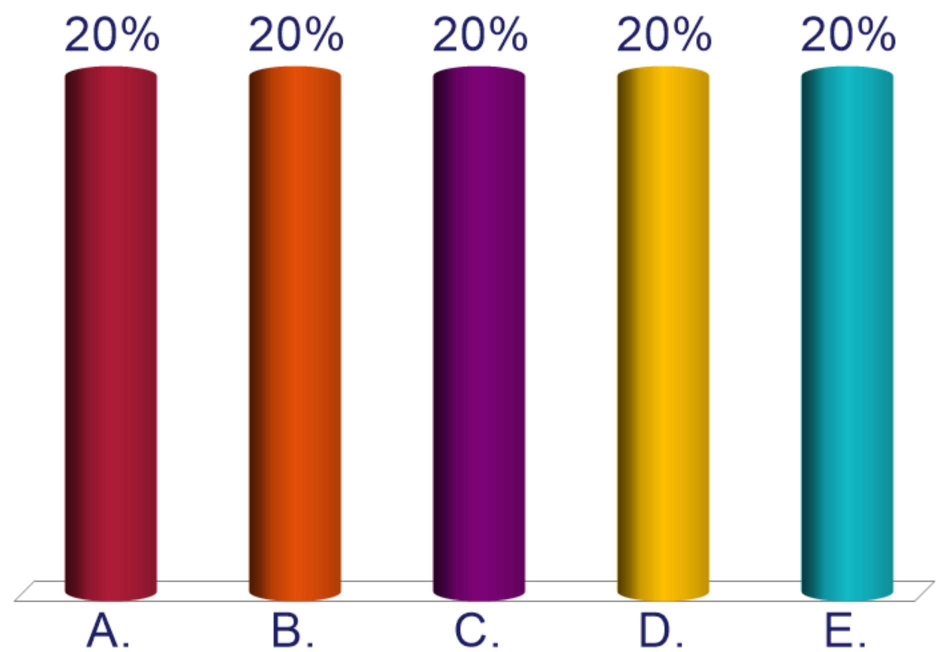
- What is muscle tissue?
- Function of muscle tissue
- Types of muscle tissue
    - Smooth
    - Cardiac
    - Skeletal
- Morphological classification of muscle tissue
    - Striated
    - Smooth
- Skeletal muscle
    - Location, microscopic structure
- Cardiac muscle
    - Location, microscopic structure
- Smooth muscle
    - Location, microscopic structure

Clicker Question Next!

Please prepare to answer the clicker question on the next slide.

A 34-year-old man comes to the emergency department because of a 20-minute history of pain and swelling to his thigh. He received a chop wound to his thigh 20 minutes ago. His blood pressure is 134/85 mmHg, and his pulse is 88/min. Physical examination shows a deep laceration to the right anterior thigh, which has severed some of muscle tissue in that area. Which of the following is most likely a normal microanatomical finding of the severed tissue?

A. Intercalated discs
B. Peripheral nuclei
C. Central nuclei
D. Caveolae
E. Branched

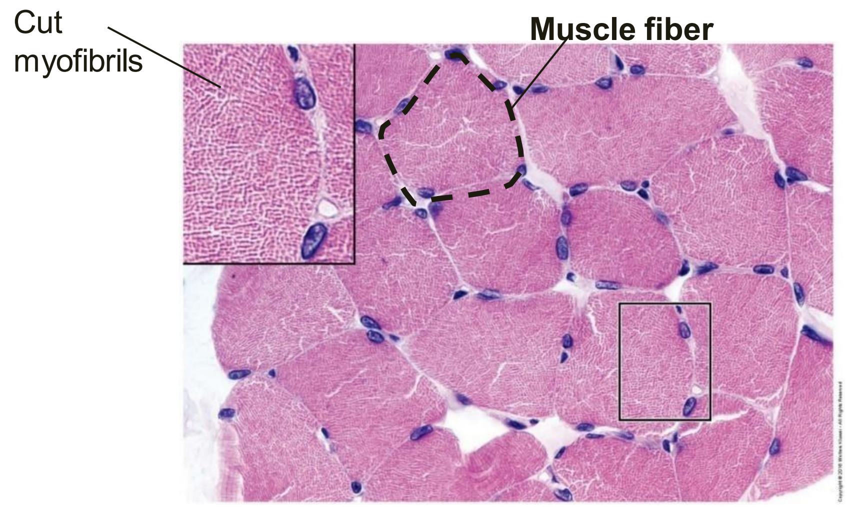


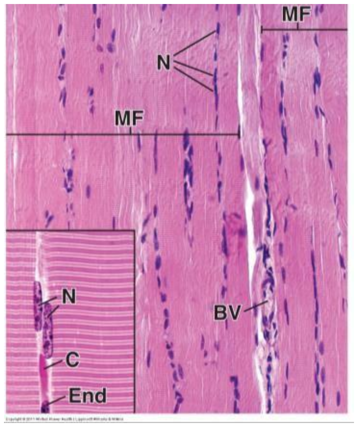
Skeletal Striated Muscle Tissue

- Usually attached to the skeletal system
- Under voluntary control
- Formed from the fusion of individual myoblasts (muscle cells)

General organization:

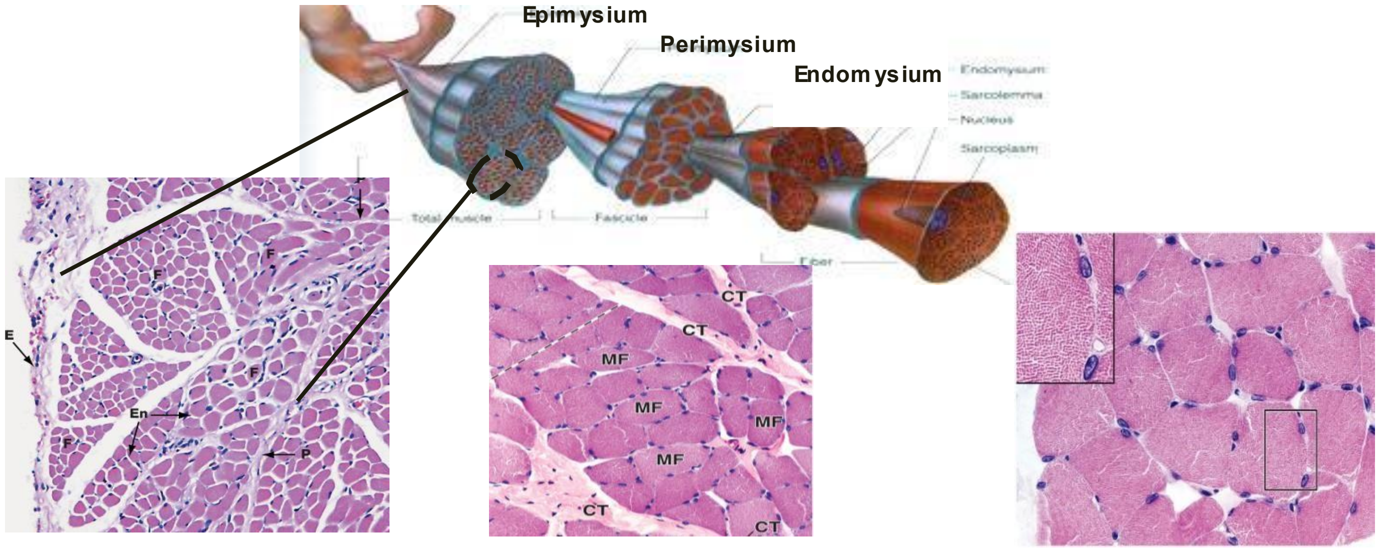
- Multinucleated with nuclei just beneath the sarcolemma
- Longitudinal sections:
  - Cylindrical
    - Uniformed length
  - Cross-sections:
    - Regular shaped
    - All cells appear roughly around the same size
- Held together by connective tissue – see next slide





Skeletal Striated Muscle Tissue

Connective tissue coverings



Epimysium- E
Dense connective tissue encasing **multiple** fascicles.

- Contains major blood vessels and nerves
- Continues with tendon to attach muscle at the *myotendinous junction*
- Absent in cardiac and smooth muscles

Perimysium- P
Groups of skeletal myocytes/fibers form a **fascicle (F)**

- Each fascicle is surrounded by a layer of connective tissue or perimysium
- Contains larger blood vessels & Nerves
  Absent in cardiac and smooth muscles

Endomysium- En
Delicate layer of reticular fibers that surrounds **individual** muscle fiber (myocyte)

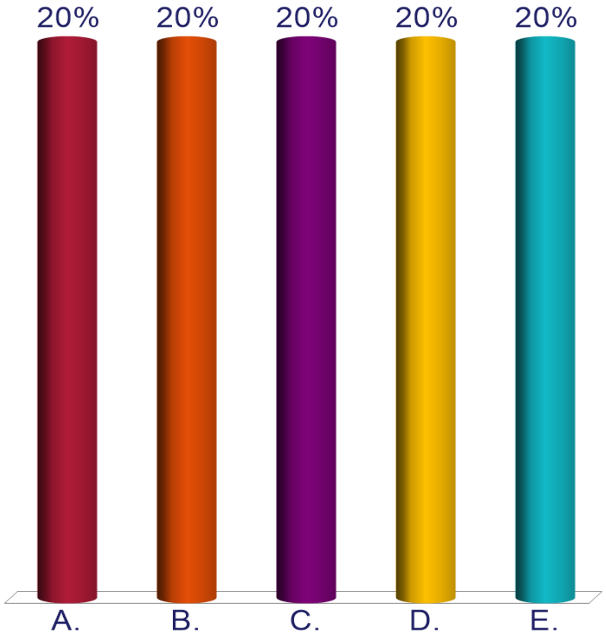
- Contains small blood vessels and very fine neuronal branches
- Found in cardiac and smooth muscles

Clicker Question Next!

Please prepare to answer the clicker question on the next slide.

A 46-year-old man comes to the emergency department because of a 2-hour history of progressive, generalized muscle weakness. He is a marine biologist and was stung by a fish during a diving excursion 4 hours ago. Physical examination shows excessive salivation and muscle weakness in all extremities. The physician suspects that the fish may have released a cytotoxin that inhibits the polymerization of actin filaments from its globular form (G-actin) to its filamentous form (F-actin). Which of the following statements best characterizes the role of the affected protein in muscle contraction?

A. Calcium binding and storage
B. Regulation of thick filament
C. Bind to thick filament
D. Inhibit thick filament's function
E. Align thick filaments in the sarcomere



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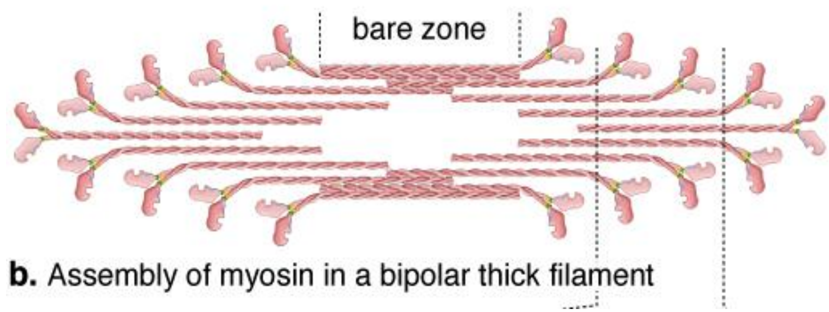
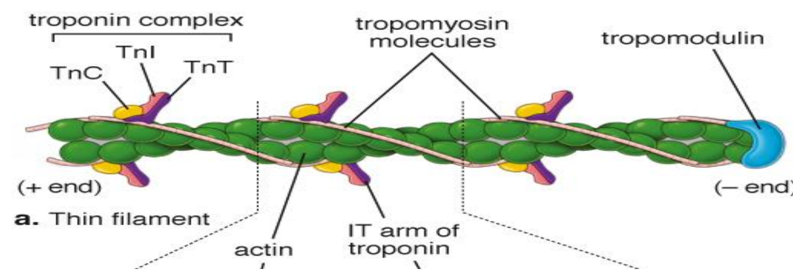
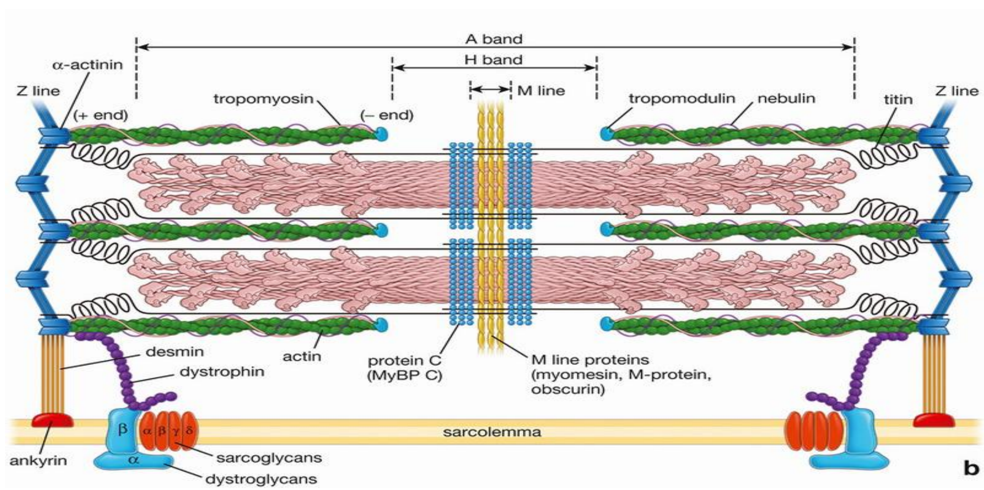
Myofilaments and the Sarcomere





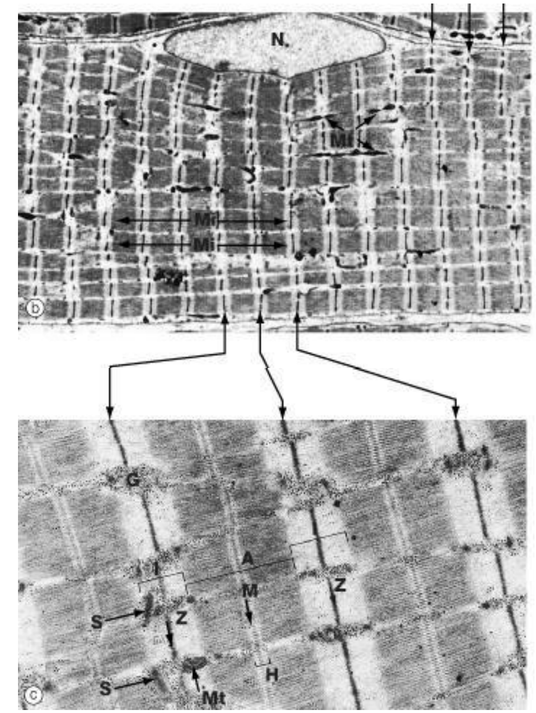


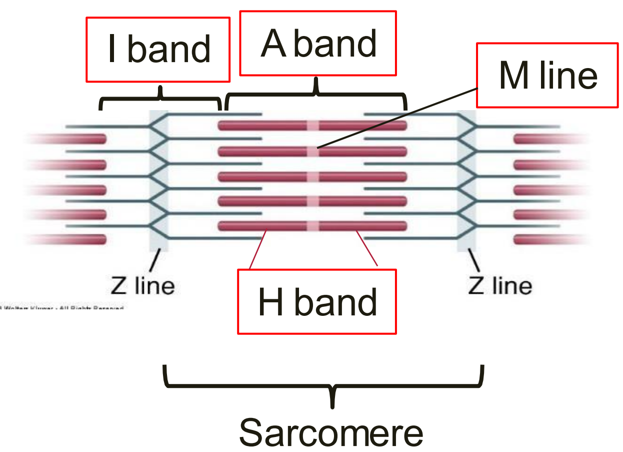
Sarcomere and Striated Muscles

- Specific arrangement of myofilaments creates dark and light bands which accounts for the cross-striation characteristics of **all** striated muscles
  - egs. skeletal and cardiac muscle
- This specific arrangement/organization forms a **sarcomere**

Sarcomere

- Basic contractile unit of striated muscles
- Contains the contractile proteins (myofilaments)
- Defined as the segment of the myofibril between two (2) adjacent Z-lines/discs
- Thick filaments (myosin II) occupy more of the center of the sarcomere
  - Forms the A-band
    - Bisected by the H-band
      - Bisected by the M-line
- Thin filaments (F-actin) are attached to the Z-line
  - Extends into the A-band to the edge of the H-band
- I-band
  - Parts of adjacent sarcomere on either side of the Z-line
  - Contains only thin (F-actin) filaments



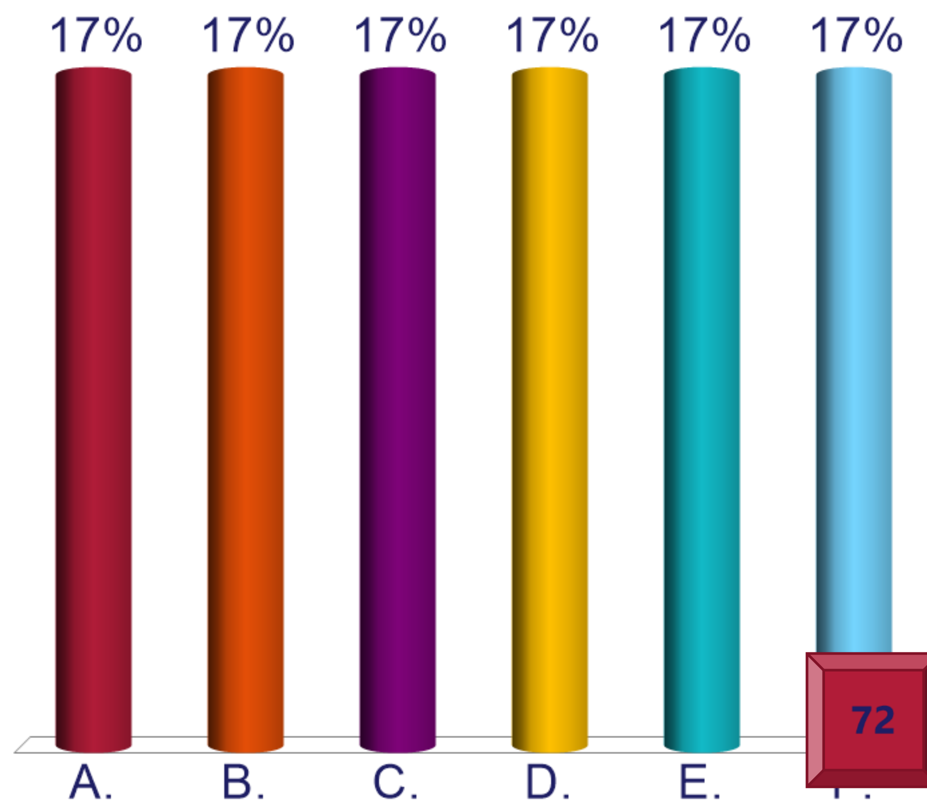


Clicker Question Next!

Please prepare to answer the clicker question on the next slide.

A 76-year-old woman is brought to the emergency department after being found unconscious in her apartment. Her vital signs are unrecordable. Physical examination shows a very stiff, cold woman with ashen skin tone. Her light pupillary reflexes are absent. She is pronounced dead and recognized to be in rigor mortis. Rigor mortis is a stage of death with sustained skeletal muscle contraction due to maintenance of the actomyosin cross bridges because of unregulated increased cytosolic calcium levels. Which of the following best identifies the reservoir of the unregulated ion in muscle fibers?

A. Mitochondria
B. Lysosomes
C. Transverse tubules
D. Terminal cisternae
E. Myofilaments
F. Troponin C



Contractile Apparatus for Striated Muscles

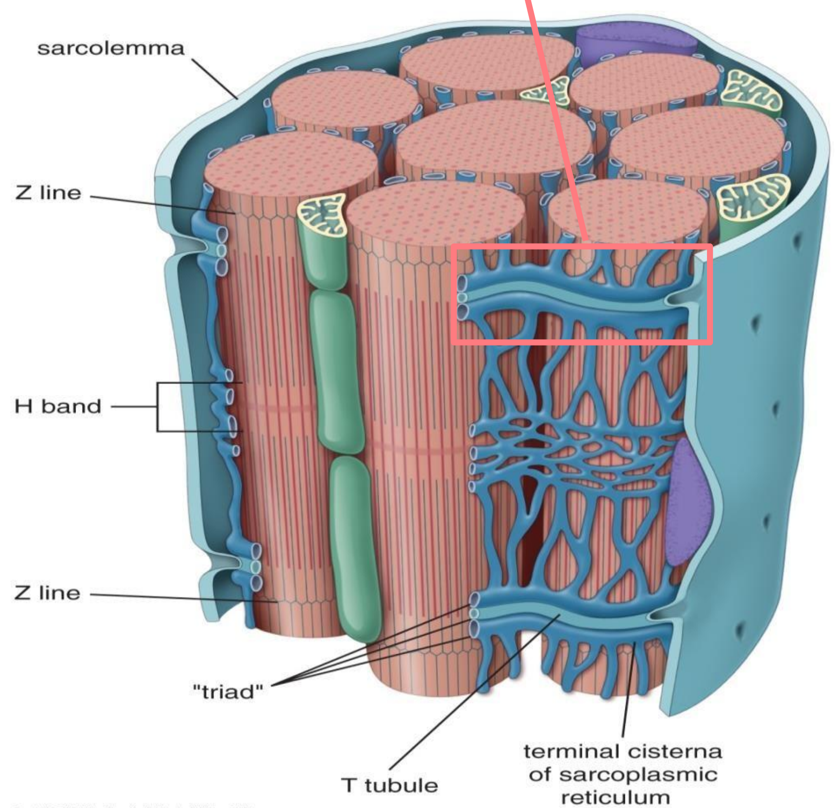
Components

- Myofilaments
  - Contractile proteins – F-actin, myosin II
- Sarcoplasmic reticulum
  - Well-developed smooth endoplasmic reticulum
    - Membranous compartment of flattened sacs
  - Forms a tubular network around the myofilaments
  - Functions in the rapid delivery and removal of calcium ions from the cytoplasm
  - Has dilated ends called terminal cisternae
    - Houses calcium ions
    - Communicates directly with the T-tubules
    - Location varies based on the type of striated muscle
- Transverse tubules (T-tubules)
  - Invaginations of the sarcolemma
  - Location varies based on the type of striated muscle
- Calcium ions (Ca2+)

Localization of Contractile Apparatus: Skeletal Muscle

- **Sarcoplasmic reticulum (SER)**
  - Extends from one A-I junction to the next A-I junction within the same sarcomere
- **Terminal cisternae**
  - Enlargements at the A-I junctions that houses calcium ions
- **Transverse tubules (T-tubules)**
  - Invaginations of the sarcolemma at the A-I junction
- **Triad**
  - Found **at the A-I junction**
  - 1 T-tubule and 2 terminal cisternae
  - **Helps to deliver calcium to the cell's cytoplasm**
- Mitochondria scattered between the myofibrils

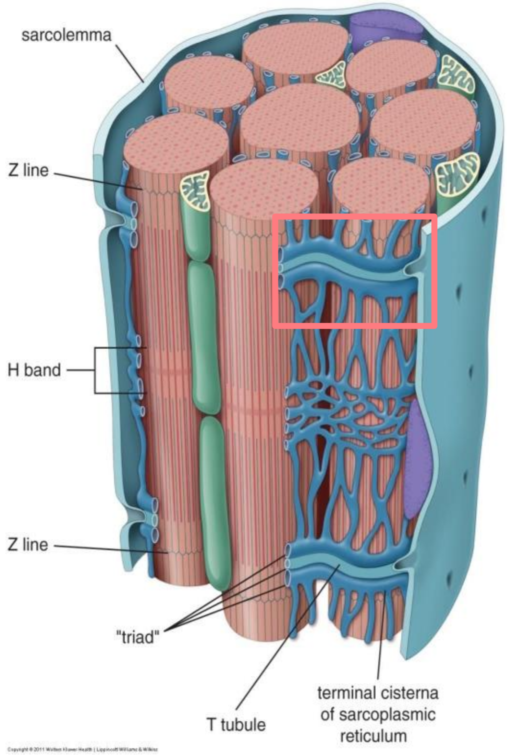
Triad (Td) : 1 T tubule + 2 terminal cisternae at the A-I junction

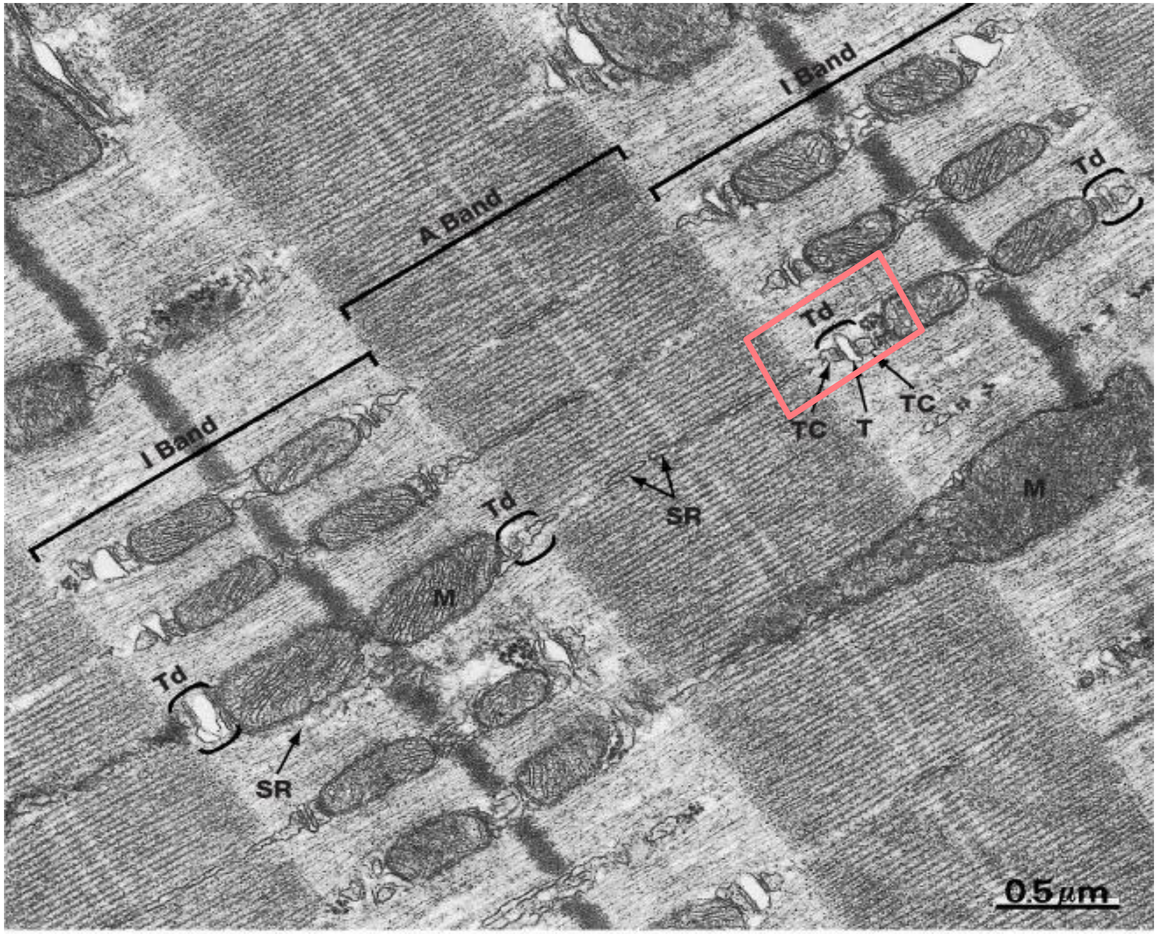


How does it work?

Depolarization of T-tubule membrane triggers release of Ca^{2+} from terminal cisterna & initiates muscle contraction cycle

Localization of Contractile Apparatus: Skeletal Muscle





Muscle Contraction

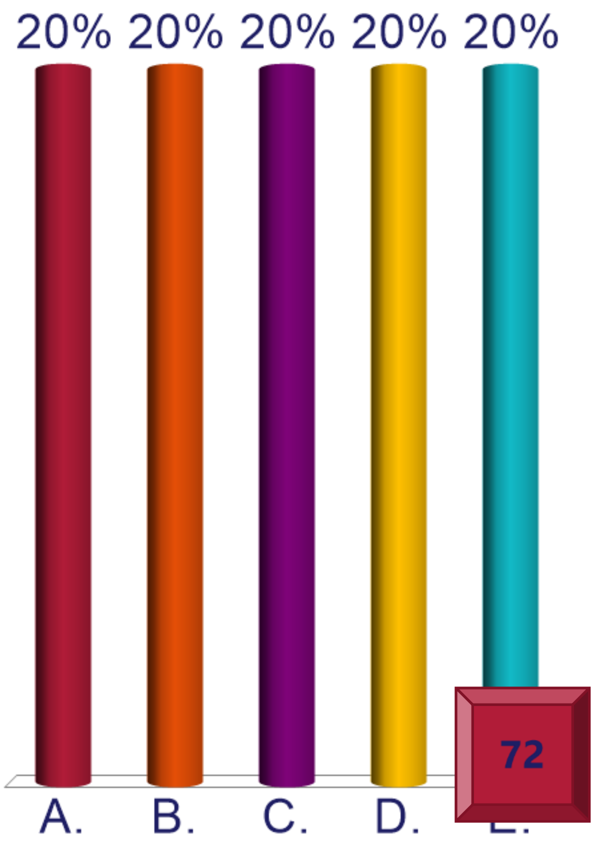
- Muscle contraction is a result of interaction between thick (myosin II) and thin (F-actin) filaments
- In skeletal muscles, many fibers contracting together results in gross movement of that particular muscle group
- **Sliding Filament Hypothesis of Huxley**
  - Sarcomere shortens and becomes thicker, but the myofilaments remain the same length
    - Myofilaments slide past each other
    - Increase the amount of overlap helps them to maintain their lengths
  - Sliding action results from repeated "make and break" attachments between the heads of the myosin molecules and neighboring actin filaments – actomyosin crossbridge
  - A band remains constant
  - I band and H band both decreases in size
  - Z lines are drawn closer to the ends of the A bands
- **Stages of the Contraction Cycle**
- Attachment, release, bending, force generation and reattachment

Clicker Question Next!

Please prepare to answer the clicker question on the next slide.

A 4-year-old boy is brought to the physician by his parents because of a 2-day history of difficulty rising from a lying position and a 3-month history of muscle pain and stiffness, a waddling gait, and frequent falls. Genetic testing confirms the near-absence of the dystrophin gene. This gene codes for the protein dystrophin. Which of the following best characterizes the function of the affected protein?

A. Anchors actin filaments to the Z line
B. Link extracellular matrix to the actin filament
C. Anchors the Z line to the sarcolemma
D. Prevents excessive stretching of the sarcomere
E. Binds thick filaments at the M line



Accessory Proteins of Skeletal Muscles Cells' Contractile Apparatus

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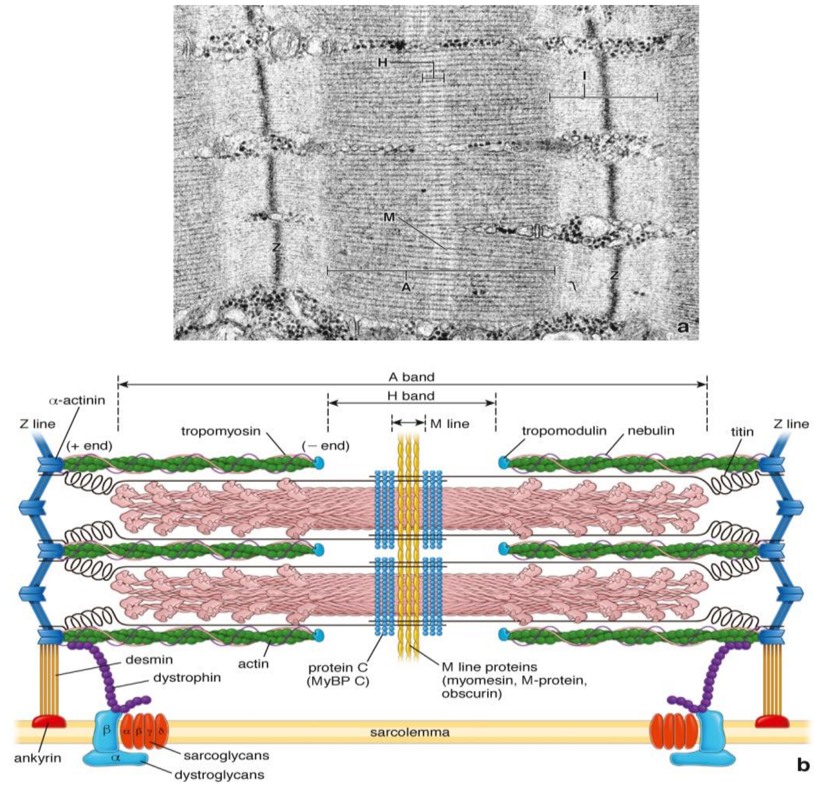
- Essential in regulating the spacing, attachment and alignment of the myofilaments within the sarcomere
- Includes:
- **Titin**
  - Extends from the Z-line toward the M line
  - "Molecular springs" – prevents excessive stretching of the sarcomere
  - Helps centre the thick filaments between the Z-lines
- **α-Actinin**
  - Bundles actin filaments in parallel array
  - Anchors actin filaments at the Z-line
- **Desmin**
  - Intermediate filament
  - Attaches Z-line to each other and to the plasma membrane
    - Uses linker protein ankyrin
- **M line proteins**
  - Binds thick filaments at the M line
  - Attach titin molecules to the thick filaments
  - Egs. Myomesin, M-protein, obscurin, muscle creatine phosphatase
- **Myosin binding protein C (MyBP-C)**
  - Stabilization of thick filaments
- **Dystrophin**
  - Link laminin to thin filaments
  - Absence results in Duchenne Muscular Dystrophy

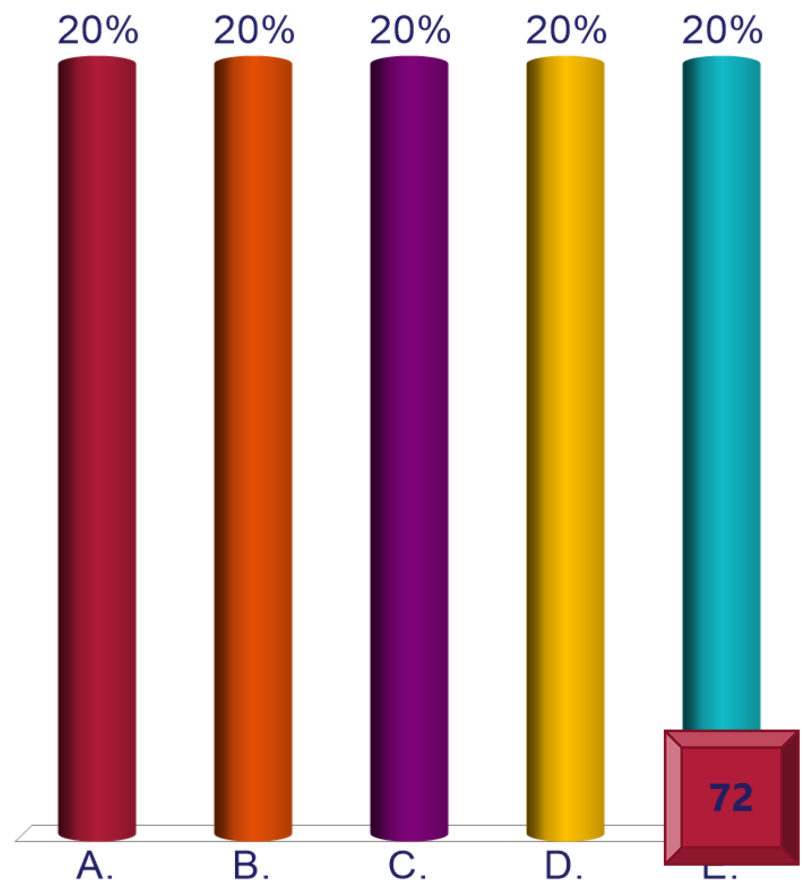


Clicker Question Next!

Please prepare to answer the clicker question on the next slide.

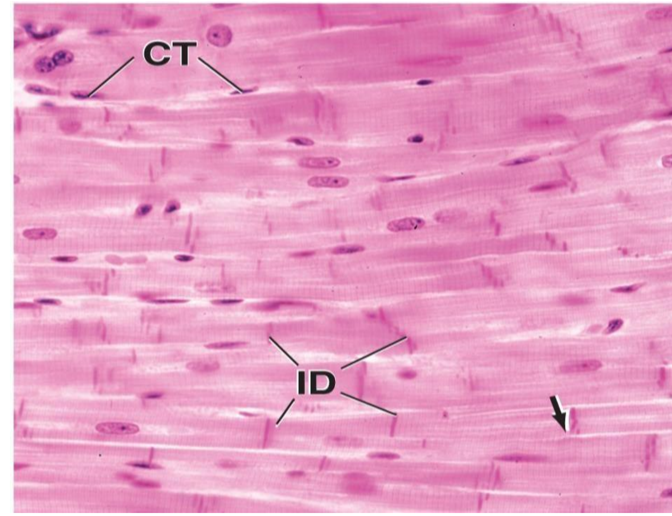
A 40-year-old man comes to the emergency department because of a 30-minute history of chest pains, and shortness of breath and a 3-hour history of dizziness. His blood pressure is 80/60 mmHg and pulse is 102/min. Physical examination shows fluid in his lungs. An ECG shows changes consistent with a myocardial infarction (heart attack). Which of the following best characterizes the infarcted tissue?

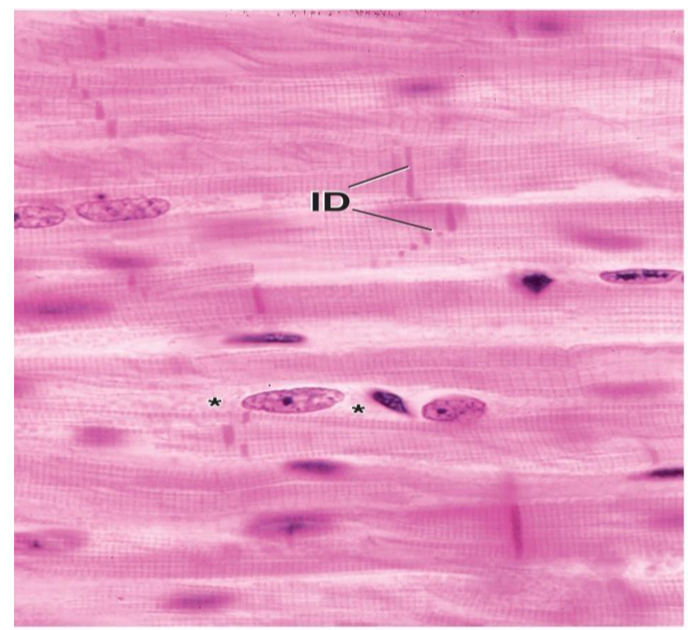
A. Single, central nucleus
B. Cylindrical and non-branching
C. Single, peripheral nucleus
D. Multiple, peripheral nucleus
E. Non-striated



Cardiac Striated Muscle Tissue

- Forms the bulk of the myocardium
- May extend slightly into the walls of some of the great vessels
- Involuntary control
- Muscle fibers (cardiac myocytes)
    - Branched
    - Surrounded by endomysium
    - Single, centrally located nuclei
    - Some may be bi-nucleated
    - Cylindrical in long axis, but is of varying lengths
    - Arranged end-to-end
        - Held in place by intercalated discs
- Perinuclear region is free of myofibrils and houses the organelles
- Numerous large mitochondria & glycogen stores
- Granules containing diuretic hormones: atrial natriuretic peptide and brain natriuretic peptide



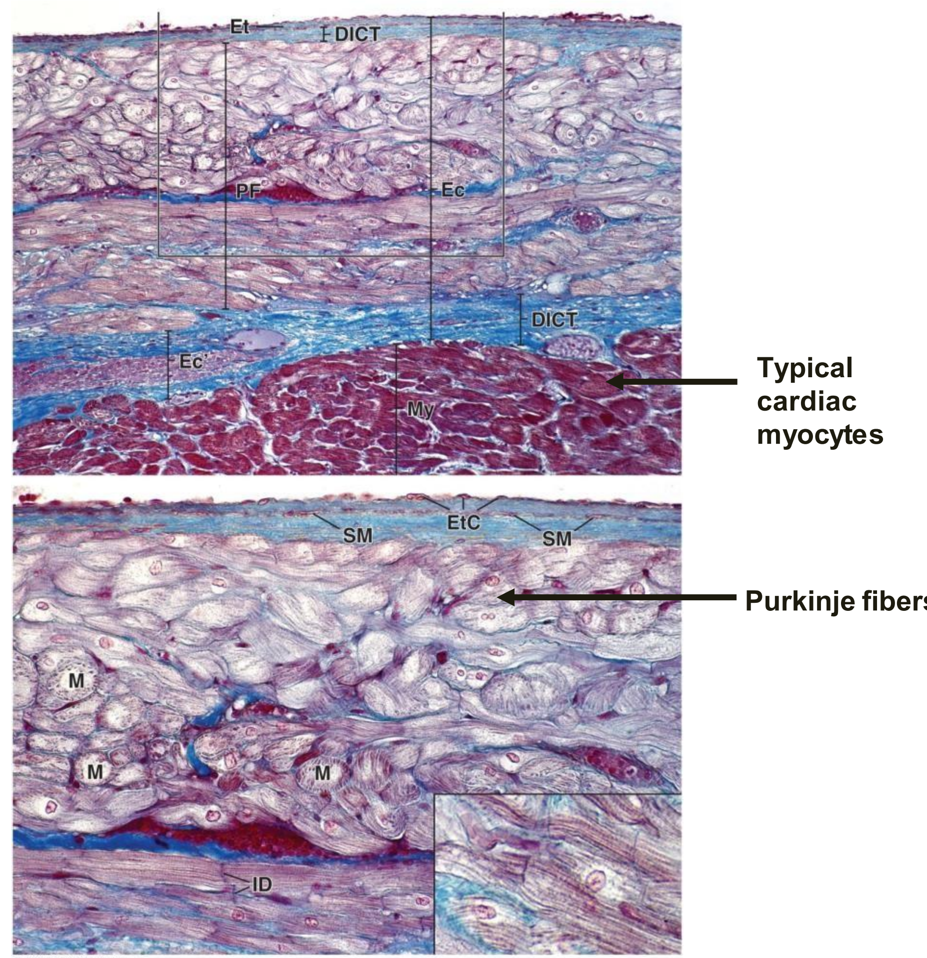


ID – intercalated discs
* - perinuclear region

Cardiac Muscle

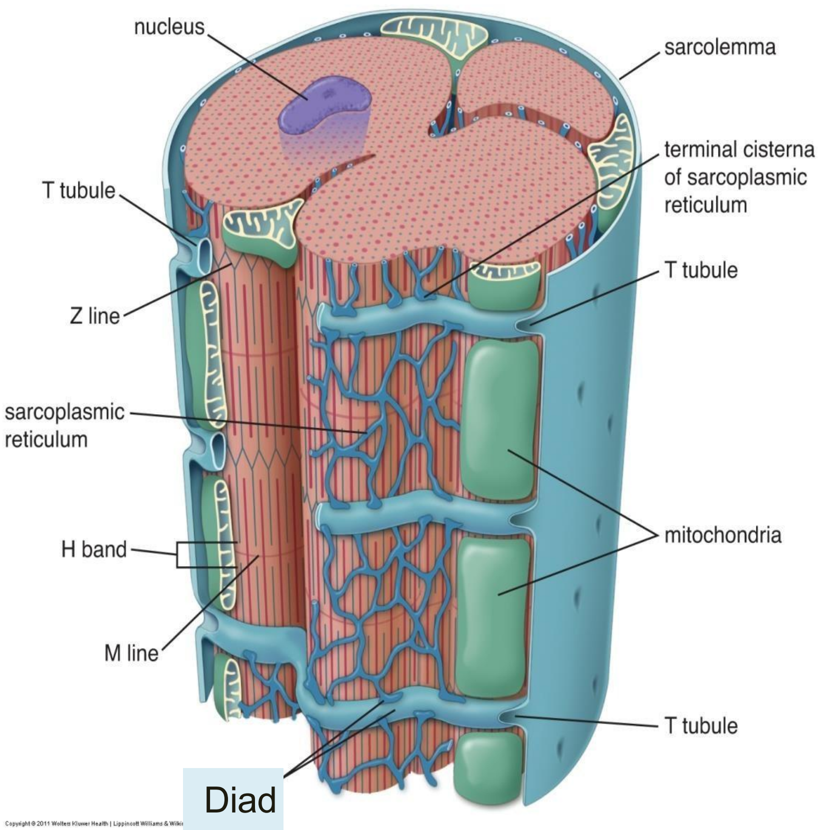
Purkinje fibers

- Large, modified cardiac muscle cells located just deep to the endocardium in the **subendocardial** connective tissue
- Specialized to conduct impulses of the A-V bundle and allow synchronization of ventricular contraction
- Pale staining due to
  - few myofibrils(these are located peripherally)
  - large amount of glycogen
- Abundant mitochondria



Localization of Contractile Apparatus: Cardiac Muscle

- **Sarcoplasmic reticulum**
  - Single network along the sarcomere
  - Extends from Z line to Z line
  - Less developed than in skeletal muscle
- **Terminal cisterna**
  - Contain Ca^{2+} release channels
    - Release Ca^{2+} into sarcoplasm
- **Transverse tubules (T-tubules)**
  - Located at the Z line (***compare this to skeletal muscle***)
- **Diad**
  - Complex of one (1) T-tubule and one (1) adjacent terminal cisternae **at the Z-line**
  - **Helps to deliver calcium to the cell's cytoplasm**



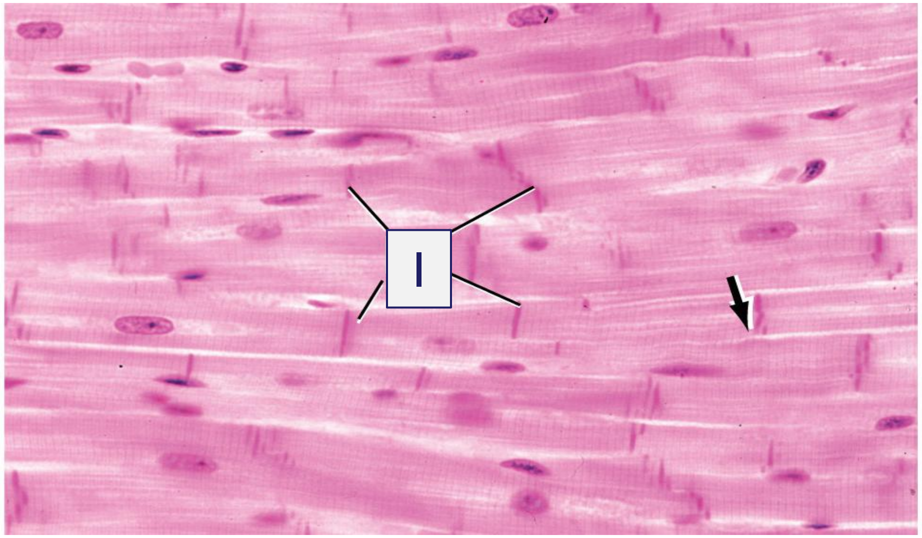
How does it work?

Depolarization of T-tubule membrane triggers release of Ca^{2+} from terminal cisterna & initiates muscle contraction cycle

Clicker Question Next!

Please prepare to answer the clicker question on the next slide.

A 19-year-old man is brought to the emergency department because of a 30-minute history of loss of consciousness. He has recently recovered from COVID-19 infection. His pulse is 132/min. An ECG tracing shows an irregular, rapid heart rate. Physical examination shows an abnormal heart-beat. The SARS-CoV-2 virus is the causative agent for COVID-19. This virus causes direct injury to the cardiac myocytes and promotes disruption to the structure identified as "I" in the attached photomicrograph. Which of the following best identifies this structure?



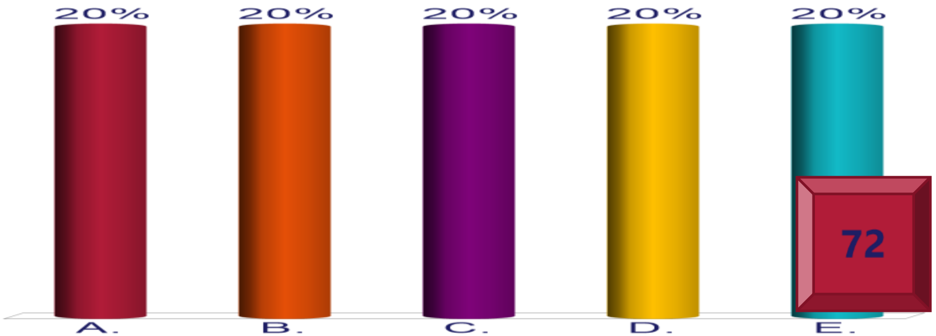
A. Caveolae

B. Sarcoplasmic reticulum

C. Intercalated discs

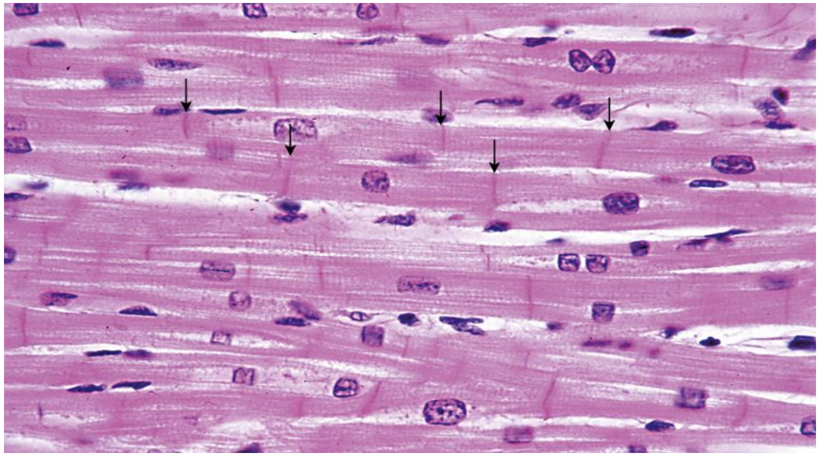
D. Terminal cisternae

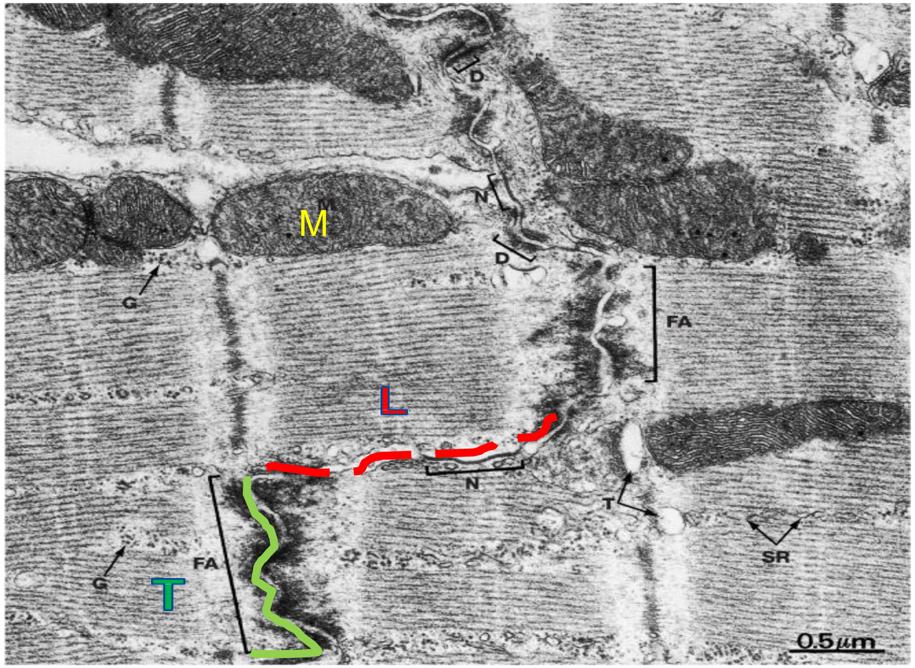
E. Sarcomere



Cardiac Muscle: Intercalated disc

- Intercalated discs are attachment sites between adjacent cardiac myocytes
- Transversely (T) oriented parts of the intercalated disk
    - seen at right angle to the myofibril like the risers of a stairway
    - Contains macula adherens and fascia adherens
- Longitudinal or lateral parts (L) are parallel to myofibril like the steps of a stairway
    - Contains gap junctions and macula adherens
- Mitochondria (M) are abundant in cardiac muscle due to the high metabolic demands of these cells





Cardiac Muscle: Intercalated disc

Transverse component (T)

(FA)Fascia adherens (adhering junctions)

- Major structural element
- Binds cardiac muscle cells at their ends
- Serves as attachment site for thin filaments in terminal sarcomeres

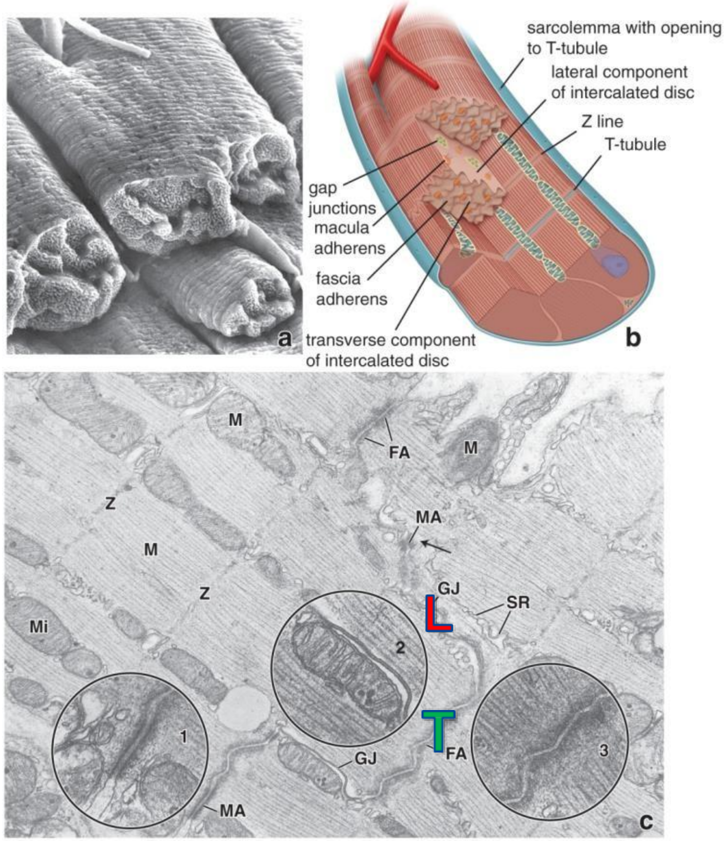
(MA) Maculae adherentes (desmosomes)

- Bind individual muscle cells to each other
- Reinforce fascia adherens
- Found in both transverse & lateral component.

Lateral component (L)

(GJ)Gap junctions (communicating junctions)

Maculae adherents (MA)

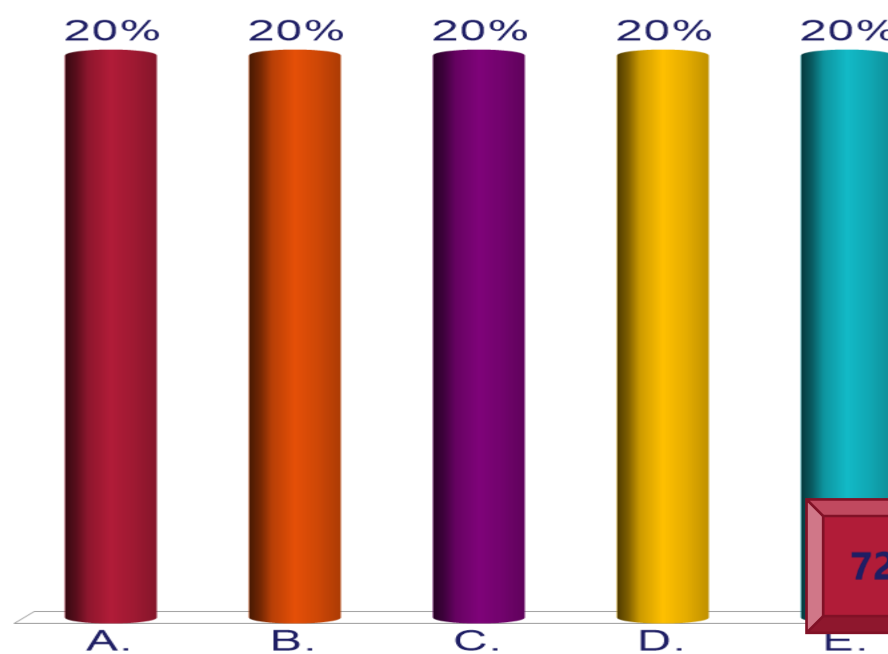


Clicker Question Next!

Please prepare to answer the clicker question on the next slide.

A 32-year-old woman comes to the physician because of a 1-day history of worsening left-sided abdominal pain. Imaging studies show a urinary stone lodged in the lumen of the left ureter. The physician explains that this results in continuous contraction of the smooth muscle cells within the wall of the affected structure. Which of the following best identifies the calcium binding protein in these cells?

A. Myosin light chain kinase
B. Tropomyosin
C. Desmin
D. Calponin
E. Calmodulin
F. Troponin C



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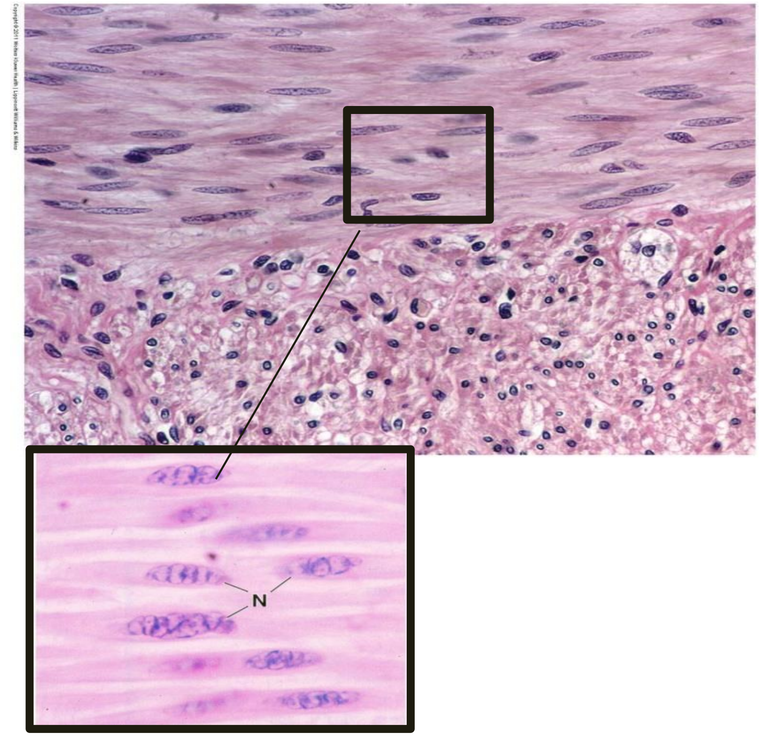
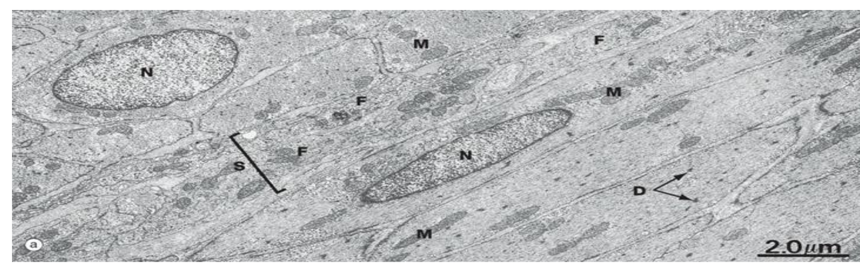
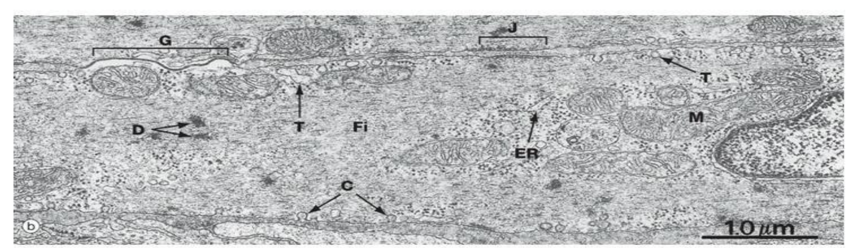
Smooth Muscle

- Elongated, fusiform cells with tapered ends
- Interconnected via gap junctions ((J) in EM below)
- Generally organized into bundles or sheets around the lumen of hollow organs e.g. gastrointestinal tract, uterus, blood vessels
- Central nucleus
  - Appears cork-screwed in contracted state
  - Appears rounded in cross section
- No cross-striations thus even staining with H&E staining
  - Lacks a sarcomere
  - Still possess contractile proteins
- Sarcoplasm
  - Pinocytotic vesicles
  - Dense bodies (D)
    - Forms attachment sites for thin (actin) filaments and intermediate filament (desmin or vimentin)
    - Composed of plaque proteins eg α-actinin

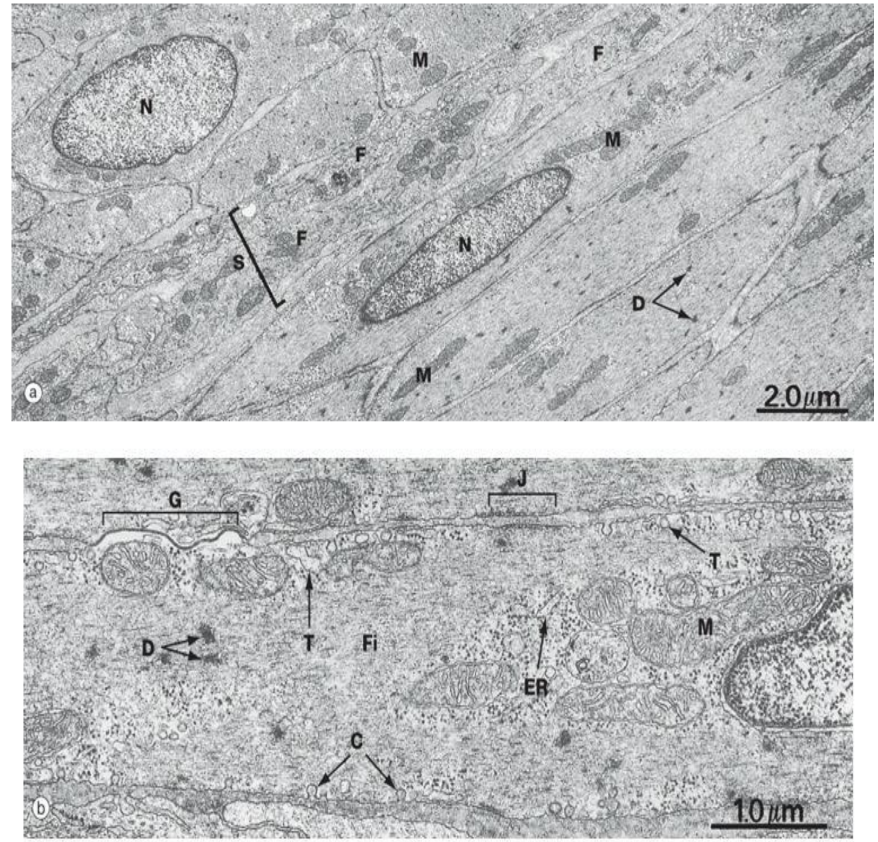






Smooth Muscle

- **Smooth muscle (SM) cells secrete connective tissue matrix**
- Synthesize both type IV collagen and type III collagen
- Elastin, proteoglycans, and multi-adhesive glycoproteins
- Vascular and uterine SM also secrete large amounts of type I collagen and elastin
- **Possess:**
  - Well-developed rER (ER) & golgi (G)

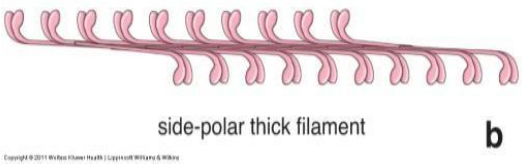


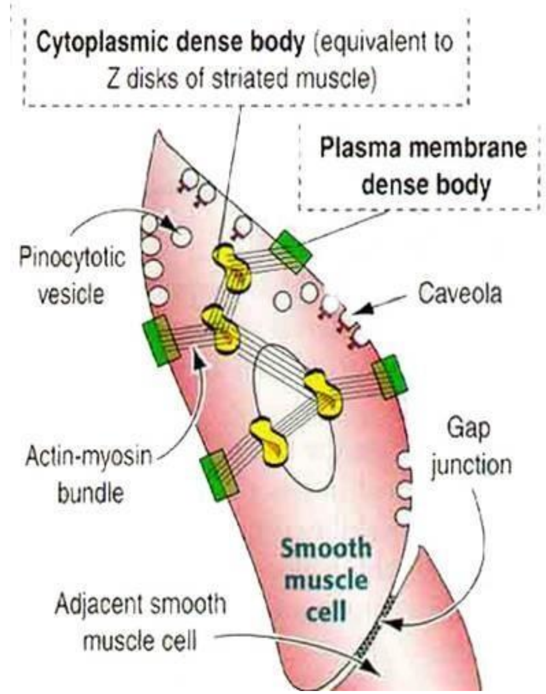
S – Sarcoplasmic reticulum
D – Dense bodies
M – Mitochondria
C - Caveolae

Contractile Apparatus of Smooth Muscle

Components:

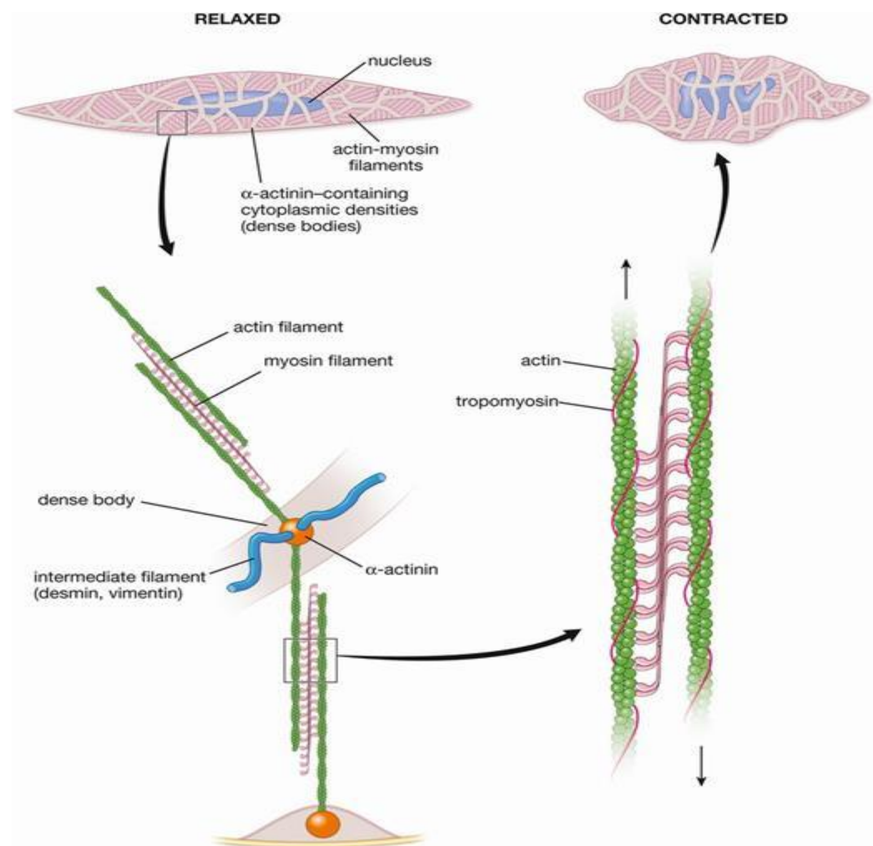
- Myofilaments
  - Actin filaments
  - Myosin filaments – smooth muscle myosin (SMM)
- Sarcoplasmic reticulum (SER)
  - Sparse profiles unlike straited muscles
- Caveolae
  - Invaginations of the sarcolemma
  - **Resembles T-tubules** of striated muscles
  - Lies close to sarcoplasmic reticulum
  - Surrounded by cytoplasmic vesicles
    - **Vesicles resemble terminal cisternae, but does not house calcium ions**
- Calcium (Ca2+) ions
- Dense bodies
  - Analogs of Z-line
  - Distributed throughout the sarcoplasm (cytoplasmic dense bodies) in a network of intermediate filaments or attached to the sarcolemma (plasma membrane dense body)
- Type III intermediate filaments (desmin, vimentin)
  - Provide link between dense bodies, cytoskeleton and sarcolemma
    - This link aids in contraction





Associated Proteins of Smooth Muscle Cells' Contractile Apparatus

- Essential in the initiation and regulation of contraction
- They include:
  - Myosin light chain kinase (MLCK)
    - initiates contraction
  - Calmodulin
    - Ca2+ binding protein
    - Regulates the intracellular calcium concentration
    - Ca^{2+} -calmodulin complex binds & activates MLCK
    - Related to TnC found in skeletal muscles
  - α-Actinin
    - Structural component of dense bodies



How does it work?

In relaxed state

- Free calcium ions are sequestered in the sarcoplasmic reticulum

Nerve stimulation:

- Free calcium is released into the sarcoplasm
  - Binds to calmodulin
    - Calcium-calmodulin complex then activates MLCK
      - Phosphorylates myosin
        - Myosin, once phosphorylated, can now binds to actin
          - Actin and myosin filaments slide past each other like as in striated muscles
- Because the myofilaments are arranged in a crisscross pattern inserted around the sarcolemma, contraction results in shortening of the muscle fiber
  - Muscle fiber assumes a globular shape in contrast to the elongated shape in the relaxed state